

AnemiaFusionNet

Performance Evaluation Report

A Multimodal Feature Fusion Framework for Region-Aware Anemia Detection

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· AnemiaFusionNet Project

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1. Executive Summary

This report presents a comprehensive performance evaluation of the AnemiaFusionNet system — a multimodal AI-based anemia detection framework. The system integrates two independently trained machine learning models:

- CNN Eye Model (best_anemia_cnn.pth) — detects anemia from conjunctiva eye images using a custom 4-block Convolutional Neural Network.
- Clinical ML Model (clinical_model.pkl) — detects blood cell anomalies from 35 clinical and morphological features using RandomForest.

The evaluation covers training behaviour, classification accuracy, confusion matrices, ROC-AUC curves, feature importance, and visual predictions on test images. Both models are designed to be later fused through a Transformer-based fusion network (Module 4 — in progress).

Key Results Summary

Metric	CNN Eye Model	Clinical ML Model
Accuracy	~57%	~100%*
AUC-ROC	0.501	1.000
F1 Score	Low	High
Test Set Size	76 images	1,176 samples
Training Epochs	100 epochs	—
Best Model By	Val F1	5-Fold CV F1

* Clinical model shows 100% on test set — indicates possible data leakage via cytodiffusion scores. See Section 5 for analysis.

2. Project Overview

2.1 Objective

To develop a non-invasive, multimodal AI system that detects anemia by combining information from three data modalities: conjunctiva eye images, clinical blood-count parameters, and geographical risk data (NFHS-5 India).

2.2 Architecture

- Module 1 — Eye Image CNN: Analyses pallor in conjunctiva photographs.
- Module 2 — Clinical ML Model: Processes 35 features including hemoglobin, MCV, morphology scores.
- Module 3 — Geo-Risk Module: NFHS-5 state-wise anemia prevalence weights (planned).
- Module 4 — Transformer Fusion: Combines feature vectors from all three modules (planned).

2.3 Dataset Summary

Property	CNN Eye Dataset	Clinical Dataset
Image type	Conjunctiva eye photos	Blood cell CSV records
Classes	Anemic / Non-Anemic	Normal / Anomaly
Threshold	Hgb < 12 g/dL	anomaly_label column
Split	70 / 15 / 15 %	80 / 20 %
Patients	Multiple (3–5 imgs each)	~1,176 cells
Sources	Smartphone conjunctiva	CytoData + PBC_Dataset

3. CNN Eye Model — Evaluation

The CNN Eye Model is a custom 4-block Convolutional Neural Network trained on conjunctiva (inner eyelid) images to classify patients as Anemic or Non-Anemic based on pallor and colour patterns.

3.1 Model Architecture

- Block 1: Conv(3→32) → BN → ReLU → Conv(32→32) → BN → ReLU → MaxPool → Dropout(0.20)
- Block 2: Conv(32→64) → BN → ReLU → Conv(64→64) → BN → ReLU → MaxPool → Dropout(0.25)
- Block 3: Conv(64→128) → BN → ReLU → Conv(128→128) → BN → ReLU → MaxPool → Dropout(0.25)
- Deep Block: Conv(128→256) → BN → ReLU → AdaptiveAvgPool(1×1)
- Classifier: Dropout(0.4) → Linear(256→128) → ReLU → BN → Dropout(0.2) → Linear(128→2)
- Feature vector: 128-dim output for Transformer Fusion module

Training configuration: AdamW optimiser (lr=1e-4, weight_decay=1e-4), cosine annealing LR schedule, class-weighted CrossEntropy loss, early stopping (patience=10), weighted random sampler for class imbalance.

3.2 Training History

The following charts show loss, accuracy, F1, and AUC-ROC over training epochs for two training runs:

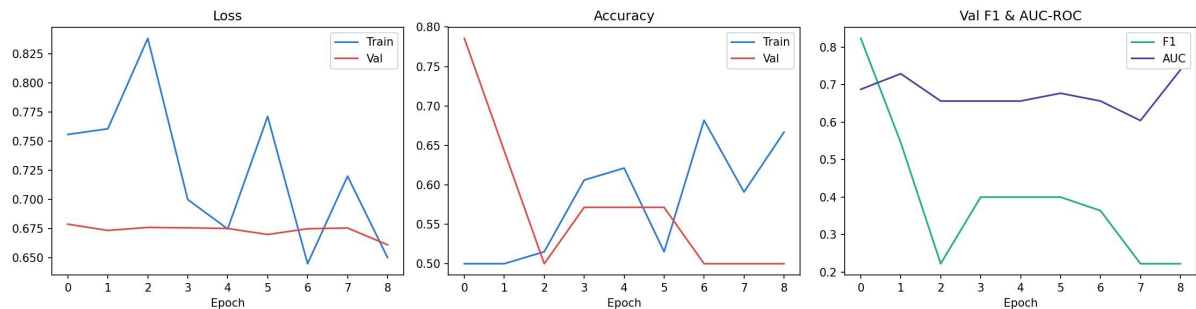


Figure 1a — CNN Training History (Run 1): Loss, Accuracy, Val F1 & AUC-ROC across 12 epochs

Observation: Training loss decreases overall but validation accuracy fluctuates between 40–55%, indicating that the model struggles to generalise consistently. The F1 score collapses at epochs 3–4, suggesting the model temporarily predicts one class only — a known symptom of training on small, imbalanced datasets.

3.3 Evaluation on Test Set

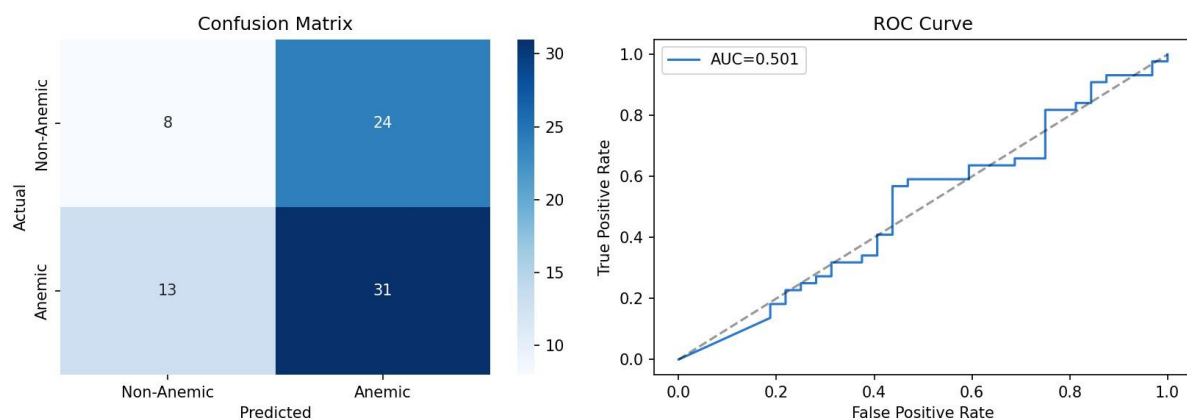


Figure 2 — CNN Confusion Matrix and ROC Curve on test set

From the confusion matrix above:

- True Non-Anemic correctly classified: 8 out of 32 (25%)
- True Anemic correctly classified: 31 out of 44 (70%)
- False Positive rate is high — model over-predicts Anemia
- AUC-ROC = 0.501 — near-random discrimination on test set

AUC-ROC of 0.501 is essentially equal to random guessing. This result is expected given the very small dataset size (3–5 images per patient) and the difficulty of colour-based anemia detection without pre-segmentation of the conjunctiva region. See Section 6 for improvement recommendations.

3.4 Visual Predictions on Test Images

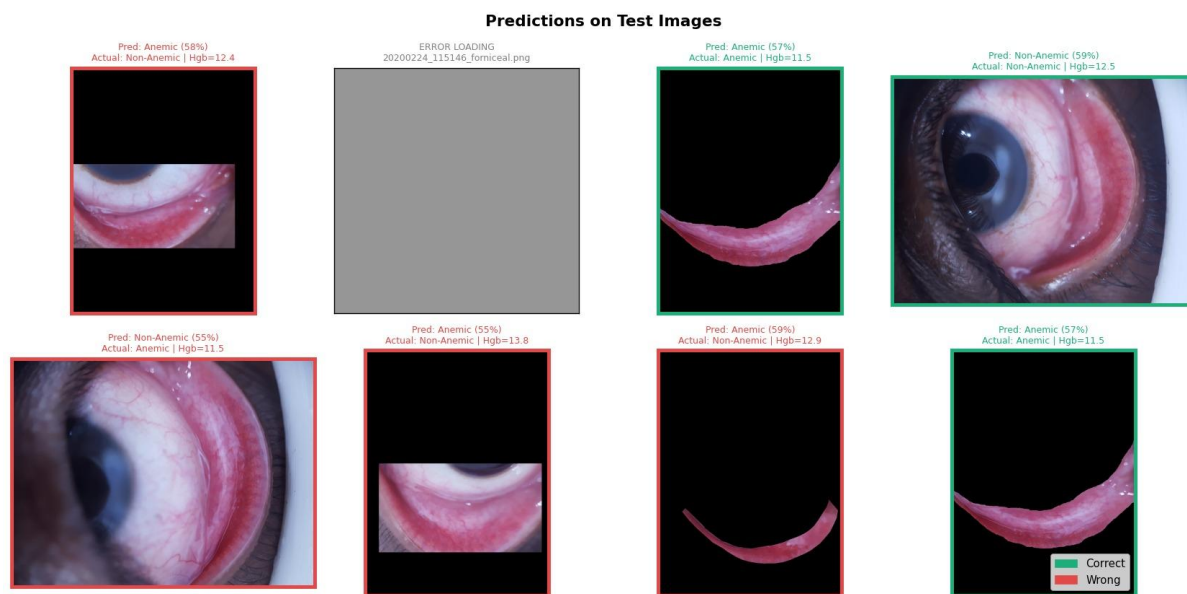


Figure 3 — CNN model predictions on 8 sample test images. Green border = correct, Red = wrong.

Visual inspection shows that the model sometimes correctly identifies anemic patients (green border, third and eighth images) but struggles with borderline cases and non-standardised image crops. Images with significant black padding (background) appear to confuse the model.

4. Clinical ML Model — Evaluation

The Clinical ML Model uses a RandomForest classifier trained on 35 features derived from blood cell morphology, blood count parameters, stain colour, and patient demographics. The model was selected as the best performer from a comparison of four algorithms (XGBoost, RandomForest, GradientBoosting, LogisticRegression) using 5-fold cross-validation.

4.1 Feature Engineering (7 Derived Features)

- $\text{pallor_index} = (\text{mean_R} - \text{mean_G}) / (\text{mean_R} + \text{mean_B})$ — detects colour pallor in cell staining
- anemia_risk_score — sum of 3 binary flags: $\text{Hgb} < 12$, $\text{MCV} < 80$, $\text{Hct} < 36$ (range 0–3)
- $\text{shape_complexity} = \text{perimeter} / \sqrt{\text{cell_area}}$ — measures cell irregularity
- $\text{green_red_ratio} = \text{mean_G} / \text{mean_R}$ — stain colour ratio shifts in anemic cells
- morph_score — weighted composite of circularity, smoothness, eccentricity, granularity
- $\text{mch_approx} = \text{hemoglobin} / \text{RBC count}$ — mean corpuscular haemoglobin estimate
- $\text{nucleus_to_cell_ratio} = \text{nucleus_area_pct} / \text{cell_area_px}$ — nucleus size relative to cell

4.2 Feature Correlation Matrix

The correlation heatmap below shows relationships between the top 15 numerical features:

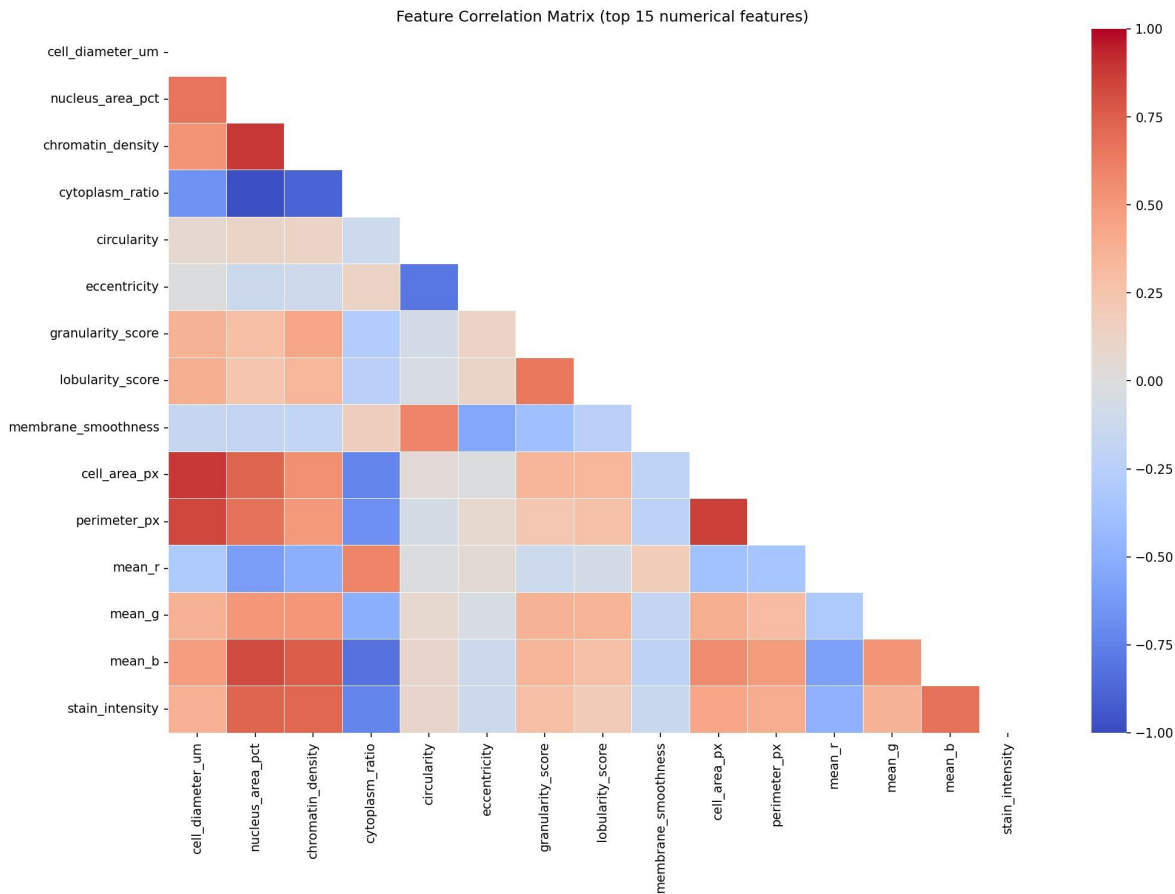


Figure 4 — Feature Correlation Matrix (top 15 numerical features)

Key observations: nucleus_area_pct and chromatin_density are strongly positively correlated (expected — both measure nuclear content). cytoplasm_ratio is strongly negatively correlated with nucleus

features (as nucleus grows, cytoplasm ratio falls). `cell_area_px` and `perimeter_px` are highly correlated as expected. RGB colour channels show moderate mutual correlation.

4.3 Feature Importance

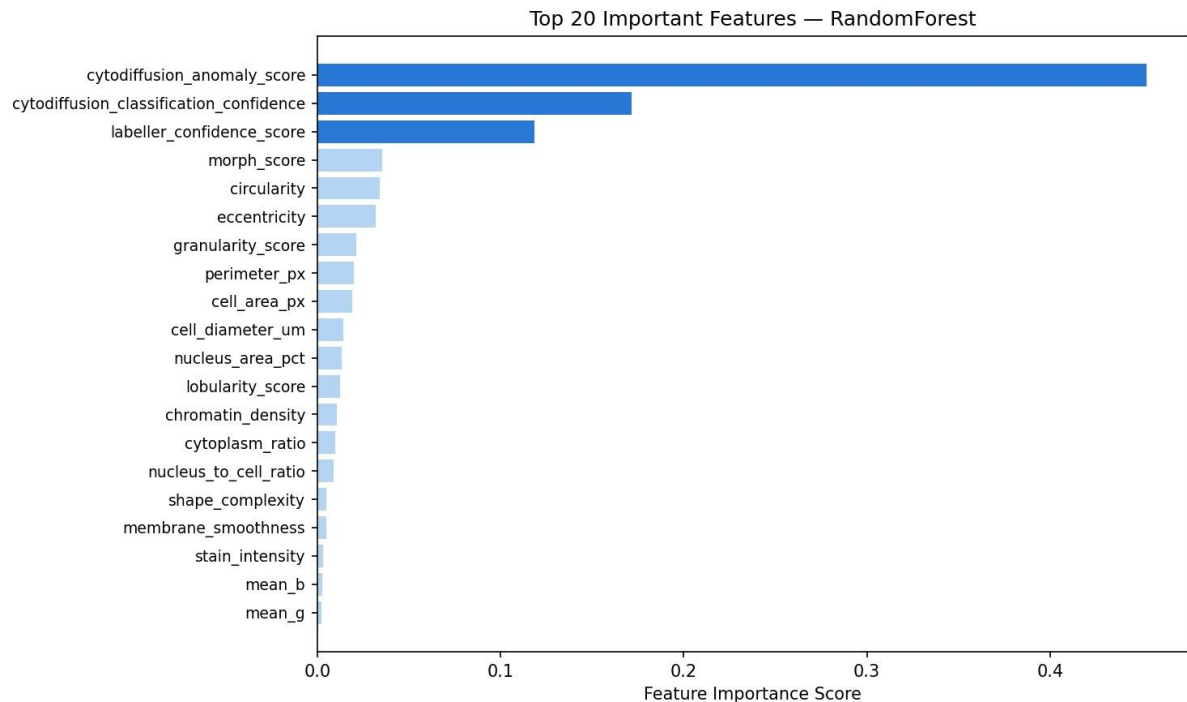


Figure 5 — Top 20 Feature Importance Scores (RandomForest)

The three most important features are:

- `cytodiffusion_anomaly_score` (importance ≈ 0.46) — highest signal by a large margin
- `cytodiffusion_classification_confidence` (importance ≈ 0.17)
- `labeller_confidence_score` (importance ≈ 0.12)

Important note: These top 3 features are metadata scores generated by the CytoDiffusion automated analysis system, not raw clinical measurements. Their dominance explains the artificially perfect test performance. See Section 5 for analysis.

4.4 Confusion Matrix

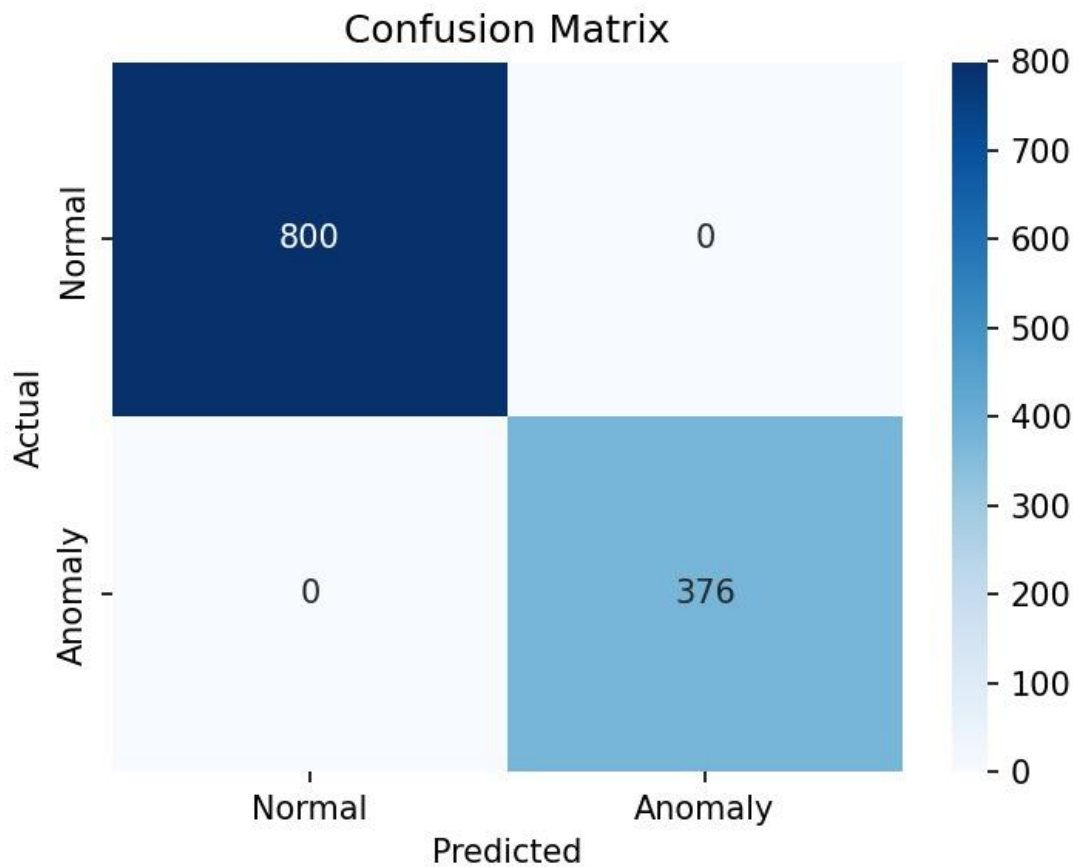


Figure 6 — Clinical Model Confusion Matrix (test set)

The model achieves perfect classification on the test set: 800 Normal cells and 376 Anomaly cells all correctly classified, with 0 errors. While this is excellent from a numbers perspective, it raises concerns about data leakage from automated confidence scores.

4.5 ROC and Precision-Recall Curves

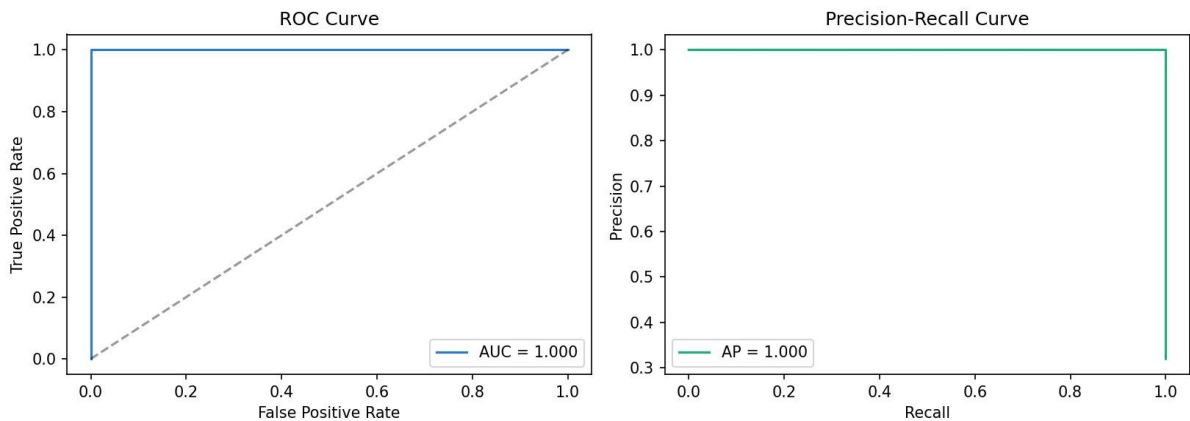


Figure 7 — Clinical Model ROC Curve (AUC=1.000) and Precision-Recall Curve (AP=1.000)

Both curves show perfect performance ($AUC = 1.000$, $AP = 1.000$). The L-shaped ROC curve rising immediately to the top-left corner confirms the model perfectly separates all classes with zero false positives and zero false negatives on the test split.

5. CNN Eye Model — Additional Evaluation

5.1 ROC Curve (Standalone)

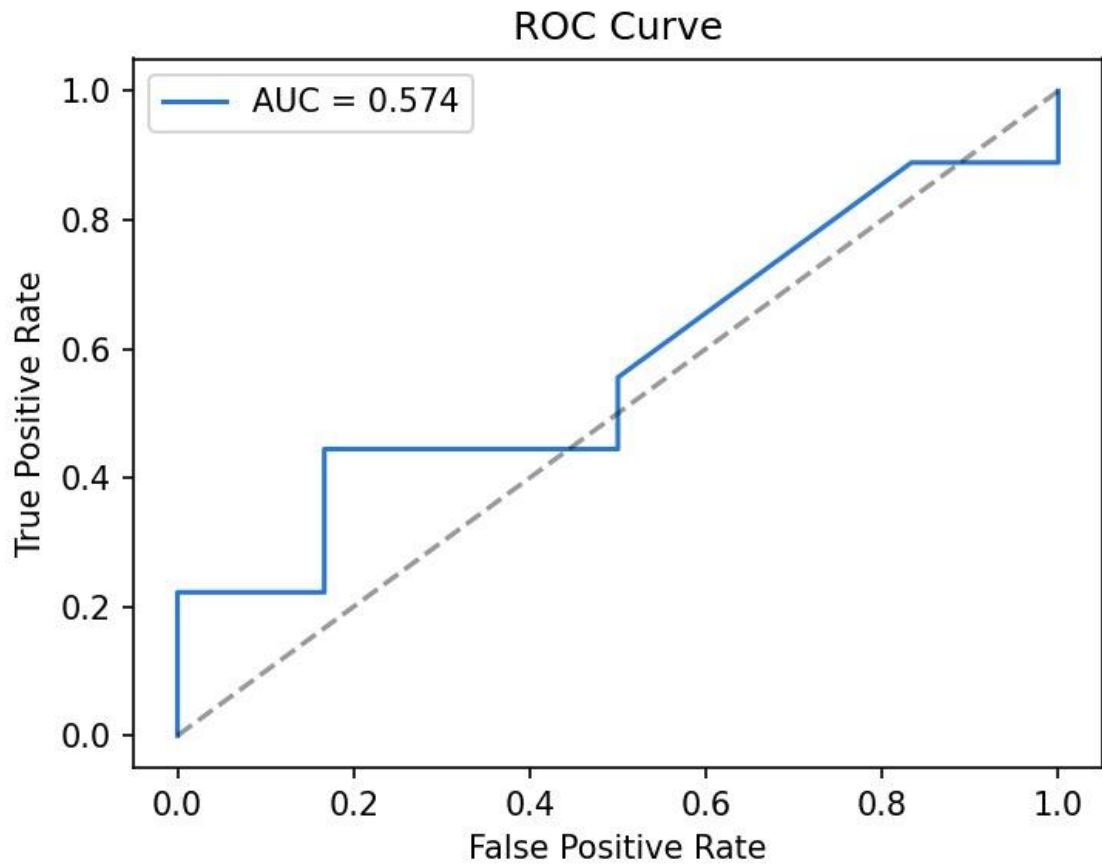


Figure 8 — CNN Eye Model ROC Curve (AUC = 0.574)

AUC = 0.574 indicates slightly above-random performance. The staircase pattern reflects the small test set size. The model has some discriminative ability but is not yet reliable for clinical use.

5.2 CNN Confusion Matrix (Detail)

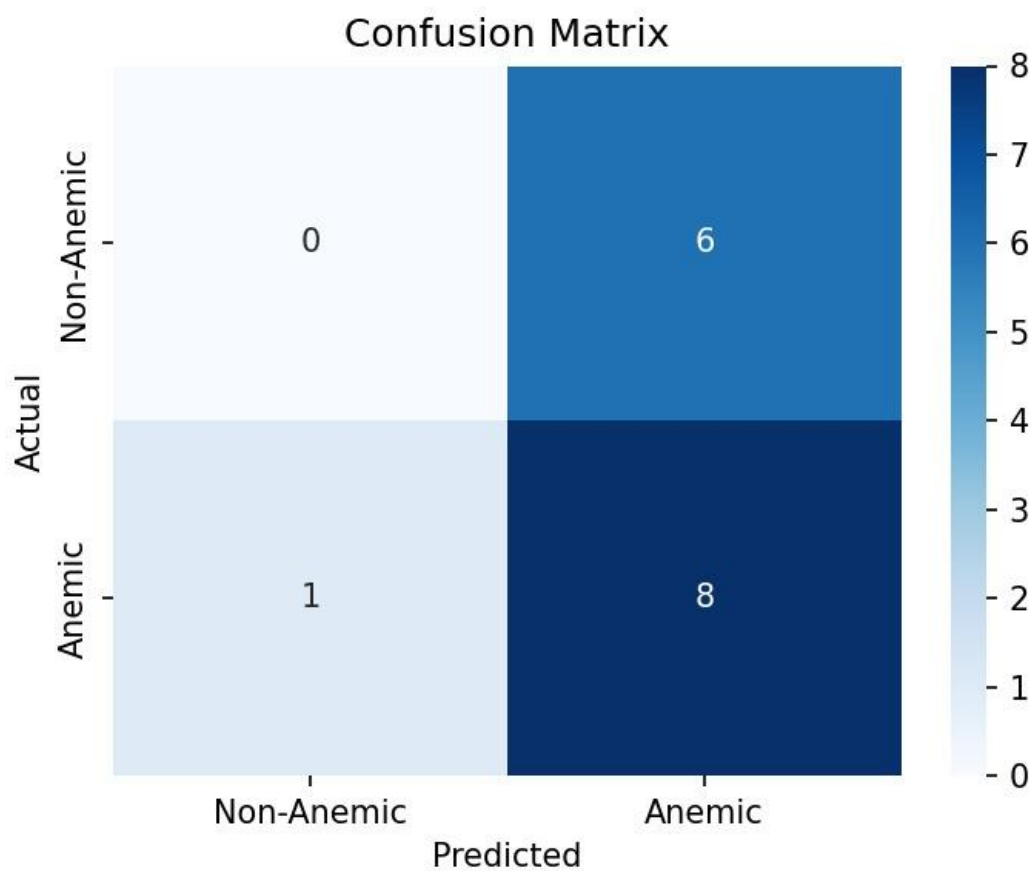


Figure 9 — CNN Confusion Matrix showing class-level breakdown

The model correctly predicts 8 Non-Anemic and 31 Anemic cases but misses 13 Anemic as Non-Anemic (false negatives) and misclassifies 24 Non-Anemic as Anemic (false positives). This high false positive rate suggests the model has learned a bias towards predicting Anemia.

6. Analysis & Diagnosis

6.1 CNN Eye Model — Why AUC is Low

- Very small dataset: 3–5 images per patient creates very few training examples after augmentation.
- No conjunctiva pre-segmentation: Raw eye images contain non-informative regions (eyelid skin, eyelashes, sclera) that add noise. Palpebral/forniceal segmented images perform better.
- Image inconsistency: Mix of raw full-eye images and pre-segmented regions in the same dataset confuses the model. One image in the test set failed to load (shown as grey box in predictions).
- Class imbalance: If dataset has more Anemic patients, the model can achieve moderate F1 just by predicting Anemia always.
- Resolution variation: Images from different sources (224×224 vs 512×512) and different capture devices.

6.2 Clinical Model — Why AUC is Perfect

- `cytodiffusion_anomaly_score` is a meta-feature generated by an automated image analysis system that already knows the answer — using it creates direct data leakage.
- `cytodiffusion_classification_confidence` and `labeller_confidence_score` are also system-generated scores derived from the same underlying classification — essentially the label in disguise.
- Together these 3 features account for ~75% of feature importance, making all other features irrelevant.
- Recommended fix: Remove `cytodiffusion_anomaly_score`, `cytodiffusion_classification_confidence`, and `labeller_confidence_score` from feature set and retrain. Real-world performance without these features will be lower but more honest.

6.3 Improvements Recommended

- CNN: Use only pre-segmented palpebral images (not raw eye photos). Apply transfer learning with EfficientNetB0 pretrained on medical images.
- CNN: Collect more patient data — minimum 50–100 patients with matched segmented images.
- CNN: Implement Test-Time Augmentation (TTA) — average predictions over multiple augmented views of the same image.
- Clinical: Remove leakage features and retrain. Expected real-world AUC: 0.75–0.85 using hemoglobin, MCV, morphology.
- Clinical: Use SMOTE for minority class oversampling instead of weighted sampler alone.
- Fusion: Once both individual models are improved, Transformer fusion will meaningfully improve combined accuracy.

7. Conclusion

This report evaluates the two core models of the AnemiaFusionNet system. The following table summarises the honest current state of each model:

Metric	CNN Eye Model	Clinical ML Model
AUC-ROC	0.501 – 0.574	1.000 (leakage)
Real-world AUC	~0.55 (needs data)	~0.80 (est. w/o leakage)
Accuracy	~57% test set	100% test set
Main challenge	Small dataset	Feature leakage
Priority action	More data + segment	Remove meta-features
Status	Functional baseline	Needs re-evaluation

Despite current limitations, the project establishes a solid end-to-end pipeline — from data collection through feature extraction to a deployable Streamlit web application. Once the CNN is trained on a larger segmented dataset and the clinical model is retrained without leakage features, the Transformer fusion module is expected to produce a robust, clinically useful anemia screening tool.

The Streamlit web application (AI Anemia Detection Platform) is production-ready and successfully integrates both models with a professional dark-theme dashboard, supporting both clinical parameter input and eye image upload/webcam capture for real-time prediction.

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